

Adaptive Filters

“We should adjust the indicators,” said Tom rigidly.

Adaptive filters can have several different meanings. For example, Perry Kaufman’s adaptive moving average (KAMA)¹ and Tushar Chande’s variable index dynamic average (VIDYA)² adapt to changes in volatility. By definition, these filters are reactive to price changes, and therefore they close the barn door after the horse is gone. The adaptive filters discussed in this chapter are the familiar Stochastic, relative strength index (RSI), commodity channel index (CCI), and band-pass filter. The key parameter in each case is the lookback period used to calculate the indicator. This lookback period is commonly a fixed value. However, since the measured cycle period is changing, as we have seen in previous chapters, it makes sense to adapt these indicators to the measured cycle period. When tradable market cycles are observed, they tend to persist for a short while. Therefore, by tuning the indicators to the measure cycle period they are optimized for current conditions and can even have predictive characteristics.

The dominant cycle period is measured using the autocorrelation periodogram algorithm. That dominant cycle dynamically sets the lookback period for the indicators. I employ my own streamlined computation for the indicators that provide smoother and easier to interpret outputs than traditional methods. Further, the indicator codes have been modified to remove the effects of spectral dilation. This basically creates a whole new set of indicators for your trading arsenal.

■ Adaptive Relative Strength Index (RSI)

Welles Wilder had just introduced the RSI³ when I began trading. I asked my broker why 14 days was used in the calculation. His response was something like “because Wilder said so.” You have no idea how unsatisfying

that response was to an engineer, and that started me on my decades-long quest to find the optimum indicator settings. I think I can now describe those optimum settings.

Wilder's original RSI was defined by

$$RSI = 100 - \left[\frac{100}{1 + RS} \right]$$

where

$$RS = \frac{\text{Average of 14 day's closes UP}}{\text{Average of 14 day's closes DOWN}}$$

"Closes UP" meant that if today's close was higher than yesterday's close, we would add the difference in closing prices to the average. Similarly, "closes DOWN" meant that if today's close was lower than yesterday's, we would add the positive difference in closing prices to the average. We can simplify the RSI equation if we substitute $RSI = CU / CD$. Then,

$$RSI = 100 - \left[\frac{100}{1 + \frac{CU}{CD}} \right] = 100 - \left[\frac{100 * CD}{CU + CD} \right] = \frac{100 * CU + 100 * CD - 100 * CD}{CU + CD}$$

$$RSI = \frac{100 * CU}{CU + CD}$$

Increasing generality, CU can mean the sum of closes up over any interval, and CD can mean the sum of closes down over the same period. Additionally, the factor of 100 is recognized as providing a simple scaling. So, RSI is just the ratio of the sum of closes up to the sum of the absolute value of all the differential closing prices.

$$RSI = \frac{CU}{CU + CD}$$

The question then becomes what is the best length to use for the RSI summation. If we use an extremely long length, then the closes up will be approximately half of the absolute value of all the closing price differentials. In this case, the RSI would hover near 0.5 and would be basically worthless. However, if the length used for summation is very short, then the RSI would be stuck at 1 for extended periods in uptrends and would be stuck at 0 for extended periods in downtrends. In a perfect world where prices swing as a

sine wave having the period of the dominant cycle, the RSI would just touch the value of 1 if the upswing half of the dominant cycle is used for the summation. Similarly, the RSI would just touch the value of 0 if the downswing half of the dominant cycle is used for the summation. Therefore, to get the optimum swing of the RSI indicator, the best adaptive summation length to use is half the measured dominant cycle.

The adaptive RSI starts with the computation of the dominant cycle using the autocorrelation periodogram approach. Rather than describing the computations, the reader is referred to the description in Chapter 8. Code Listing 11-1 contains the EasyLanguage code for the adaptive RSI, and the identification of the RSI indicator itself following the dominant cycle calculation is noted by the comment near the end. Since the objective is to use only those frequency components passed by the roofing filter, the variable *Filt* is used as a data input rather than closing prices. Rather than independently taking the averages of the numerator and denominator, I chose to perform smoothing on the ratio using the SuperSmoother filter described in Chapter 3. The coefficients for the SuperSmoother filters have previously been computed in the dominant cycle measurement part of the code. I have included reference lines at the 30 percent and 70 percent levels, but these can easily be adjusted to 20 percent/80 percent or whatever else serves the purpose.

Code Listing 11-1. EasyLanguage Code for the Adaptive RSI

```
{
    Adaptive RSI
    (c) 2013 John F. Ehlers
}

Vars:
    AvgLength(3),
    M(0),
    N(0),
    X(0),
    Y(0),
    alpha1(0),
    HP(0),
```

(Continued)

```

a1(0),
b1(0),
c1(0),
c2(0),
c3(0),
Filt(0),
Lag(0),
count(0),
Sx(0),
Sy(0),
Sxx(0),
Syy(0),
Sxy(0),
Period(0),
Sp(0),
Spx(0),
MaxPwr(0),
DominantCycle(0),
Color1(0),
Color2(0),
Color3(0);

Arrays:
    Corr[48](0),
    CosinePart[48](0),
    SinePart[48](0),
    SqSum[48](0),
    R[48, 2](0),
    Pwr[48](0);

//Highpass filter cyclic components whose periods are shorter
than 48 bars
alpha1 = (Cosine(.707*360 / 48) + Sine (.707*360 / 48) - 1) /
Cosine(.707*360 / 48);
HP = (1 - alpha1 / 2)*(1 - alpha1 / 2)*(Close - 2*Close[1] +
Close[2]) + 2*(1 - alpha1)*HP[1] - (1 - alpha1)*(1 -
alpha1)*HP[2];
//Smooth with a Super Smoother Filter from equation 3-3
a1 = expvalue(-1.414*3.14159 / 10);

```

```

b1 = 2*a1*Cosine(1.414*180 / 10);
c2 = b1;
c3 = -a1*a1;
c1 = 1 - c2 - c3;
Filt = c1*(HP + HP[1]) / 2 + c2*Filt[1] + c3*Filt[2];

//Pearson correlation for each value of lag
For Lag = 0 to 48 Begin
    //Set the averaging length as M
    M = AvgLength;
    If AvgLength = 0 Then M = Lag;
    Sx = 0;
    Sy = 0;
    Sxx = 0;
    Syy = 0;
    Sxy = 0;
    For count = 0 to M - 1 Begin
        X = Filt[count];
        Y = Filt[Lag + count];
        Sx = Sx + X;
        Sy = Sy + Y;
        Sxx = Sxx + X*X;
        Sxy = Sxy + X*Y;
        Syy = Syy + Y*Y;
    End;
    If (M*Sxx - Sx*Sx)*(M*Syy - Sy*Sy) > 0 Then Corr[Lag] =
        (M*Sxy - Sx*Sy)/SquareRoot((M*Sxx - Sx*Sx)*(M*Syy -
        Sy*Sy));
End;

For Period = 10 to 48 Begin
    CosinePart[Period] = 0;
    SinePart[Period] = 0;
    For N = 3 to 48 Begin
        CosinePart[Period] = CosinePart[Period] +
            Corr[N]*Cosine(360*N / Period);
        SinePart[Period] = SinePart[Period] + Corr[N]*Sine(360*N /
            Period);
    End;
End;

```

(Continued)

Code Listing 11-1. (Continued)

```
SqSum[Period] = CosinePart[Period]*CosinePart[Period] +
SinePart[Period]*SinePart[Period];
End;
For Period = 10 to 48 Begin
    R[Period, 2] = R[Period, 1];
    R[Period, 1] = .2*SqSum[Period]*SqSum[Period] +
        .8*R[Period, 2];
End;

//Find Maximum Power Level for Normalization
MaxPwr = .991*MaxPwr;
For Period = 10 to 48 Begin
    If R[Period, 1] > MaxPwr Then MaxPwr = R[Period, 1];
End;
For Period = 3 to 48 Begin
    Pwr[Period] = R[Period, 1] / MaxPwr;
End;

//Compute the dominant cycle using the CG of the spectrum
Spx = 0;
Sp = 0;
For Period = 10 to 48 Begin
    If Pwr[Period] >= .5 Then Begin
        Spx = Spx + Period*Pwr[Period];
        Sp = Sp + Pwr[Period];
    End;
End;
If Sp <> 0 Then DominantCycle = Spx / Sp;
If DominantCycle < 10 Then DominantCycle = 10;
If DominantCycle > 48 Then DominantCycle = 48;
//Adaptive RSI starts here, using half the measured dominant
cycle for tuning

Vars:
    ClosesUp(0),
    ClosesDn(0),
    Denom(0),
    MyRSI(0);

ClosesUp = 0;
ClosesDn = 0;
```

Code Listing 11-1. (Continued)

```

For count = 0 to IntPortion(DominantCycle / 2 - 1) Begin
    If Filt[count] > Filt[count + 1] Then ClosesUp = ClosesUp +
        (Filt[count] - Filt[count + 1]);
    If Filt[count] < Filt[count + 1] Then ClosesDn = ClosesDn +
        (Filt[count + 1] - Filt[count]);
End;
Denom = ClosesUp + ClosesDn;
If Denom <> 0 and Denom[1] <> 0 Then MyRSI = c1*(ClosesUp /
    Denom + ClosesUp[1] / Denom[1]) / 2 + c2*MyRSI[1] +
    c3*MyRSI[2];

Plot1(MyRSI);
Plot2(.7);
Plot6(.3);

```

The adaptive RSI has been applied to Dollar General (symbol DG) over roughly the last year in Figure 11.1. Note that the indicator reaches zero or one only when the prices have had a cyclic swing over half the period of the dominant cycle. These events happen relatively rarely, and show that the indicator is properly tuned.

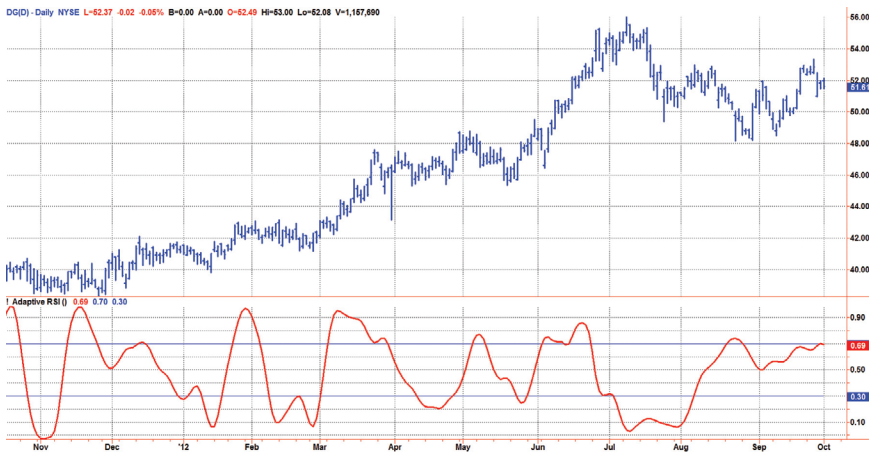


FIGURE 11.1 Adaptive RSI Is Properly Tuned to Half the Dominant Cycle Period

■ Adaptive Stochastic Indicator

The basic idea of the stochastic indicator is that it measures the current closing price relative to the lowest close over the observation range, and normalize that relative price to the range between the highest close and the lowest close over the observation range. Since finding the highest close and lowest close over the dominant cycle period is assured only by searching over the entire dominant cycle period, the entire dominant cycle period is used for the observation range in the calculations.

The adaptive stochastic indicator starts with the computation of the dominant cycle using the autocorrelation periodogram approach. Rather than describing the computations, the reader is referred to the description in Chapter 8. Code Listing 11-2 contains the EasyLanguage code for the adaptive stochastic indicator, and the identification of the indicator itself following the dominant cycle calculation is noted by the comment near the end. Since the objective is to use only those frequency components passed by the roofing filter, the variable *Filt* is used as a data input rather than closing prices.

The first step is to identify the highest “close” and the lowest “close” over the period of the current dominant cycle. The variable *Stoc* is computed as the ratio of the current value of *Filt* to the price range and is the unfiltered value of the adaptive stochastic. The *Stoc* variable is filtered using the SuperSmoother filter first describe in Chapter 3. Rather than independently taking the averages of the numerator and denominator, I chose to perform smoothing on the ratio *Stoc* using the SuperSmoother filter described in Chapter 3. The coefficients for the SuperSmoother filters have previously been computed in the dominant cycle measurement part of the code. I have included reference lines at the 30 percent and 70 percent levels, but these can easily be adjusted to 20 percent/80 percent or whatever else serves the purpose.

Code Listing 11-2. EasyLanguage Code for the Adaptive Stochastic Indicator

```
{
    Adaptive Stochastic
    (c) 2013 John F. Ehlers
}

Vars:
    AvgLength(3) ,
    M(0) ,
```



```

N(0),
X(0),
Y(0),
alpha1(0),
HP(0),
a1(0),
b1(0),
c1(0),
c2(0),
c3(0),
Filt(0),
Lag(0),
count(0),
Sx(0),
Sy(0),
Sxx(0),
Syy(0),
Sxy(0),
Period(0),
Sp(0),
Spx(0),
MaxPwr(0),
DominantCycle(0);

Arrays:
    Corr[48](0),
    CosinePart[48](0),
    SinePart[48](0),
    SqSum[48](0),
    R[48, 2](0),
    Pwr[48](0);

//Highpass filter cyclic components whose periods are
shorter than 48 bars
alpha1 = (Cosine(.707*360 / 48) + Sine (.707*360 / 48) - 1) /
Cosine(.707*360 / 48);
HP = (1 - alpha1 / 2)*(1 - alpha1 / 2)*(Close - 2*Close[1] +
Close[2]) + 2*(1 - alpha1)*HP[1] - (1 - alpha1)*
(1 - alpha1)*HP[2];
//Smooth with a Super Smoother Filter from equation 3-3

```

(Continued)

Code Listing 11-2. (Continued)

```
a1 = expvalue(-1.414*3.14159 / 10);
b1 = 2*a1*Cosine(1.414*180 / 10);
c2 = b1;
c3 = -a1*a1;
c1 = 1 - c2 - c3;
Filt = c1*(HP + HP[1]) / 2 + c2*Filt[1] + c3*Filt[2];

//Pearson correlation for each value of lag
For Lag = 0 to 48 Begin
    //Set the averaging length as M
    M = AvgLength;
    If AvgLength = 0 Then M = Lag;
    Sx = 0;
    Sy = 0;
    Sxx = 0;
    Syy = 0;
    Sxy = 0;
    For count = 0 to M - 1 Begin
        X = Filt[count];
        Y = Filt[Lag + count];
        Sx = Sx + X;
        Sy = Sy + Y;
        Sxx = Sxx + X*X;
        Sxy = Sxy + X*Y;
        Syy = Syy + Y*Y;
    End;
    If (M*Sxx - Sx*Sx)*(M*Syy - Sy*Sy) > 0 Then Corr[Lag] =
        (M*Sxy - Sx*Sy)/SquareRoot((M*Sxx - Sx*Sx)*(M*Syy -
        Sy*Sy));
End;

For Period = 10 to 48 Begin
    CosinePart[Period] = 0;
    SinePart[Period] = 0;
    For N = 3 to 48 Begin
        CosinePart[Period] = CosinePart[Period] +
            Corr[N]*Cosine(360*N / Period);
        SinePart[Period] = SinePart[Period] + Corr[N]*Sine(360*N /
            Period);
    End;
```

Code Listing 11-2. (Continued)

```
SqSum[Period] = CosinePart[Period]*CosinePart[Period] +
SinePart[Period]*SinePart[Period];
End;
For Period = 10 to 48 Begin
    R[Period, 2] = R[Period, 1];
    R[Period, 1] = .2*SqSum[Period]*SqSum[Period] +
    .8*R[Period, 2];
End;

//Find Maximum Power Level for Normalization
MaxPwr = .995*MaxPwr;
For Period = 10 to 48 Begin
    If R[Period, 1] > MaxPwr Then MaxPwr = R[Period, 1];
End;
For Period = 3 to 48 Begin
    Pwr[Period] = R[Period, 1] / MaxPwr;
End;

//Compute the dominant cycle using the CG of the spectrum
Spx = 0;
Sp = 0;
For Period = 10 to 48 Begin
    If Pwr[Period] >= .5 Then Begin
        Spx = Spx + Period*Pwr[Period];
        Sp = Sp + Pwr[Period];
    End;
End;
If Sp <> 0 Then DominantCycle = Spx / Sp;
If DominantCycle < 10 Then DominantCycle = 10;
If DominantCycle > 48 Then DominantCycle = 48;

//Stochastic Computation starts here
Vars:
    HighestC(0),
    LowestC(0),
    Stoc(0),
    SmoothNum(0),
    SmoothDenom(0),
    AdaptiveStochastic(0);
```

(Continued)

Code Listing 11-2. (Continued)

```
HighestC = Filt;  
LowestC = Filt;  
For count = 0 to DominantCycle - 1 Begin  
    If Filt[count] > HighestC then HighestC = Filt[count];  
    If Filt[count] < LowestC then LowestC = Filt[count];  
End;  
Stoc = (Filt - LowestC) / (HighestC - LowestC);  
AdaptiveStochastic = c1*(Stoc + Stoc[1]) / 2 +  
c2*AdaptiveStochastic[1] + c3*AdaptiveStochastic[2];  
  
Plot1(AdaptiveStochastic);  
Plot2(.7);  
Plot6(.3);
```

The adaptive Stochastic has been applied to Dollar General (symbol DG) over roughly the last year in Figure 11.2. An eyeball scan comparing the prices and the adaptive stochastic shows that this indicator accurately portrays the relative detrended prices over the length of the dominant cycle period, and has many characteristics in common with the adaptive RSI.

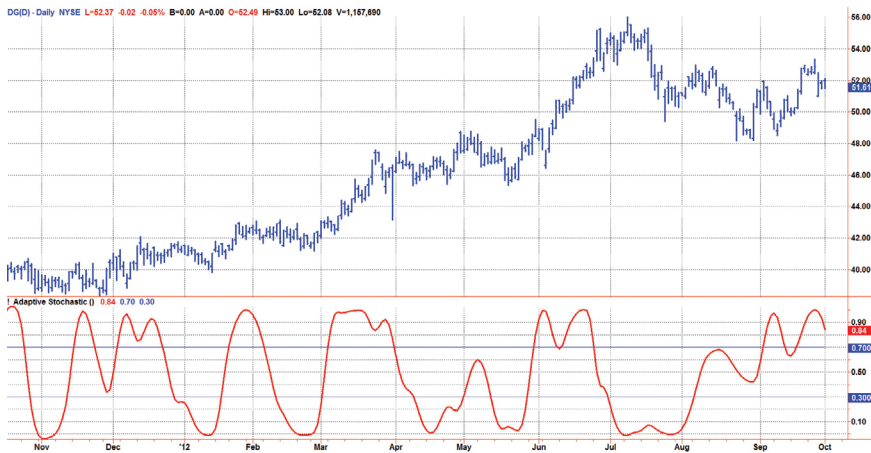


FIGURE 11.2 Adaptive Stochastic Accurately Displays the Relative Detrended Prices over the Channel Length of the Dominant Cycle Period

■ Adaptive CCI (Commodity Channel Index)

The CCI indicator was first described by Donald Lambert⁴ as a means of statistically estimating where the current prices lie within the length of a channel. He basically compares the current price relative to a moving average of prices over the length of the channel to the standard deviation of prices over the same length. A standard deviation is just the square root of the averaged squared difference of the prices from their moving averages. As a shorthand way of thinking of it, the basic idea of the CCI is taking the ratio of the “instantaneous deviation” to the standard deviation. Lambert also divides the ratio by 0.015, which is the same as multiplying the ratio by 66.7. This scaling factor sets +100 and -100 as approximations to the plus and minus one standard deviation if the data had a Gaussian probability distribution. That is not a good assumption, which can be proven heuristically by measuring the data probability distributions over a lot of data.

The time length to be used for the channel in the calculations is widely varied in the literature. In all cases, the length is rather arbitrarily established to fit the indicator to some preconceived event. It seems to me that it would be better to use one full period of the dominant cycle as the length of data to be used.

The adaptive CCI indicator starts with the computation of the dominant cycle using the autocorrelation periodogram approach. Rather than describing the computations, the reader is referred to the description in Chapter 8. Code Listing 11-3 contains the EasyLanguage code for the adaptive CCI indicator, and the identification of the indicator itself is noted by the comment near the end following the dominant cycle calculation. Since the objective is to use only those frequency components passed by the roofing filter, the variable *Filt* is used as a data input rather than the average of the high, low, and close as was done by Lambert.

The first step in the computation is to find the average “price.” It is important to use this long form to compute the average price rather than the fast algorithm of dropping off $1 / N$ of the oldest price and adding $1 / N$ of the current price to the average because N is varying from bar to bar due to the tuning using the dominant cycle period. Once the moving average is computed, it is used to compute the root mean square (RMS) of the prices. RMS is synonymous with the standard deviation. After creating the numerator and the denominator, the ratio is smoothed using the SuperSmoother filter described in Chapter 3. The coefficients for the SuperSmoother filters have previously been computed in the dominant cycle measurement part of the code. I have included reference lines at the +100 and -100 levels.

Code Listing 11-3. EasyLanguage Code for the Adaptive CCI

```
{  
    Adaptive CCI  
    (c) 2013 John F. Ehlers  
}
```

Vars:

```
AvgLength(3),  
M(0),  
N(0),  
X(0),  
Y(0),  
alpha1(0),  
HP(0),  
a1(0),  
b1(0),  
c1(0),  
c2(0),  
c3(0),  
Filt(0),  
Lag(0),  
count(0),  
Sx(0),  
Sy(0),  
Sxx(0),  
Syy(0),  
Sxy(0),  
Period(0),  
Sp(0),  
Spx(0),  
MaxPwr(0),  
DominantCycle(0);
```

Arrays:

```
Corr[48](0),  
CosinePart[48](0),  
SinePart[48](0),  
SqSum[48](0),  
R[48, 2](0),  
Pwr[48](0);
```

```

//Highpass filter cyclic components whose periods are
shorter than 48 bars
alpha1 = (Cosine(.707*360 / 48) + Sine (.707*360 / 48) - 1) /
Cosine(.707*360 / 48);
HP = (1 - alpha1 / 2)*(1 - alpha1 / 2)*(Close - 2*Close[1] +
Close[2]) + 2*(1 - alpha1)*HP[1] - (1 - alpha1)*
(1 - alpha1)*HP[2];
//Smooth with a Super Smoother Filter from equation 3-3
a1 = expvalue(-1.414*3.14159 / 10);
b1 = 2*a1*Cosine(1.414*180 / 10);
c2 = b1;
c3 = -a1*a1;
c1 = 1 - c2 - c3;
Filt = c1*(HP + HP[1]) / 2 + c2*Filt[1] + c3*Filt[2];

//Pearson correlation for each value of lag
For Lag = 0 to 48 Begin
  //Set the averaging length as M
  M = AvgLength;
  If AvgLength = 0 Then M = Lag;
  Sx = 0;
  Sy = 0;
  Sxx = 0;
  Syy = 0;
  Sxy = 0;
  For count = 0 to M - 1 Begin
    X = Filt[count];
    Y = Filt[Lag + count];
    Sx = Sx + X;
    Sy = Sy + Y;
    Sxx = Sxx + X*X;
    Sxy = Sxy + X*Y;
    Syy = Syy + Y*Y;
  End;
  If (M*Sxx - Sx*Sx)*(M*Syy - Sy*Sy) > 0 Then Corr[Lag] =
(M*Sxy - Sx*Sy)/SquareRoot((M*Sxx - Sx*Sx)*(M*Syy -
Sy*Sy));
End;

```

(Continued)

Code Listing 11-3. (Continued)

```
For Period = 10 to 48 Begin
    CosinePart[Period] = 0;
    SinePart[Period] = 0;
    For N = 3 to 48 Begin
        CosinePart[Period] = CosinePart[Period] +
            Corr[N]*Cosine(360*N / Period);
        SinePart[Period] = SinePart[Period] + Corr[N]*Sine(360*N /
            Period);
    End;
    SqSum[Period] = CosinePart[Period]*CosinePart[Period] +
        SinePart[Period]*SinePart[Period];
End;
For Period = 10 to 48 Begin
    R[Period, 2] = R[Period, 1];
    R[Period, 1] = .2*SqSum[Period]*SqSum[Period] +
        .8*R[Period, 2];
End;

//Find Maximum Power Level for Normalization
MaxPwr = .991*MaxPwr[1];
For Period = 10 to 48 Begin
    If R[Period, 1] > MaxPwr Then MaxPwr = R[Period, 1];
End;
For Period = 3 to 48 Begin
    If MaxPwr <> 0 Then Pwr[Period] = R[Period, 1] / MaxPwr;
End;

//Compute the dominant cycle using the CG of the spectrum
Spx = 0;
Sp = 0;
For Period = 10 to 48 Begin
    If Pwr[Period] >= .5 Then Begin
        Spx = Spx + Period*Pwr[Period];
        Sp = Sp + Pwr[Period];
    End;
End;
If Sp <> 0 Then DominantCycle = Spx / Sp;
If DominantCycle < 10 Then DominantCycle = 10;
If DominantCycle > 48 Then DominantCycle = 48;
```



```

Vars:
    Price(0),
    AvePrice(0),
    RMS(0),
    Num(0),
    Denom(0),
    Ratio(0),
    MyCCI(0);

//Adaptive CCI starts here, using half the measured dominant
cycle for tuning
Price = Filt;
AvePrice = 0;
For count = 0 to DominantCycle - 1 Begin;
    AvePrice = AvePrice + Price[count];
End;
AvePrice = AvePrice / DominantCycle;
RMS = 0;
For count = 0 to DominantCycle - 1 Begin;
    RMS = RMS + (Price[count] - AvePrice[count])*(Price[count] -
    AvePrice[count]);
End;
RMS = SquareRoot(RMS / DominantCycle);
Num = Price - AvePrice;
Denom = .015*RMS;
Ratio = Num / Denom;
MyCCI = c1*(Ratio + Ratio[1]) / 2 + c2*MyCCI[1] + c3*MyCCI[2];

Plot1(MyCCI);
Plot2(100);
Plot6(-100);

```

The adaptive CCI has been applied to Dollar General (symbol DG) over roughly the past year in Figure 11.3. This indicator accurately displays the level of the current prices within a channel. The duration of the channel length is determined by the measured dominant cycle period. The adaptive CCI exceeds the plus or minus one standard deviation only occasionally.

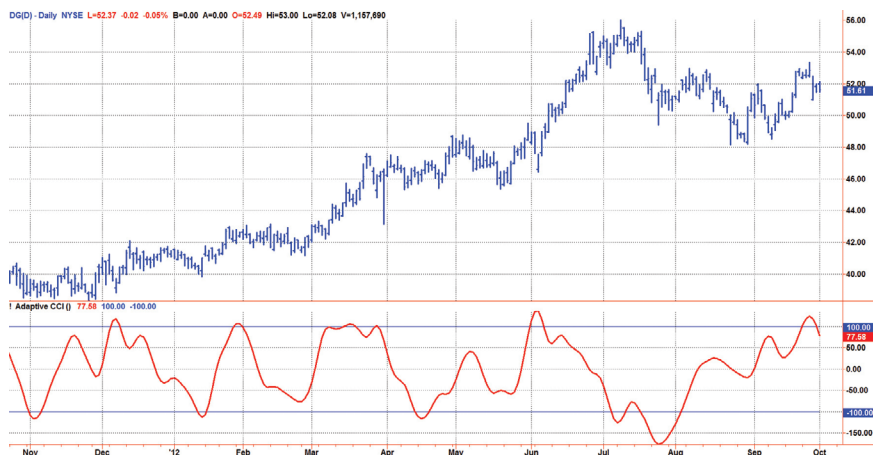


FIGURE 11.3 Adaptive CCI Shows the Price Location within the Channel Length Determined by the Measured Dominant Cycle Period

■ Adaptive Band-Pass Filter

If a band-pass filter can be designed, it just makes sense to tune that filter to the measured dominant cycle to eliminate all the other frequency components that are of no interest. The band-pass filter was described in Chapter 5. Here, the adaptive band-pass indicator starts with the computation of the dominant cycle using the autocorrelation periodogram approach. Code Listing 11-4 contains the EasyLanguage code for the adaptive band-pass indicator, and the identification of the indicator itself is noted by the comment near the end following the calculation of the dominant cycle.

One way to make a band-pass filter have a leading phase capability is to tune the filter to a period shorter than the period of the cycle being measured. In this case, the bandwidth of filter is set to 0.3. That is 30 percent of the tuned center period. Therefore, the half bandwidth is 15 percent. We tune the filter to be 10 percent toward the shorter period from the dominant cycle period to provide the phase lead while still having the data of interest be within the filter bandwidth. This provides a phase lead of the dominant cycle to be something on the order of 60 degrees, or one-sixth of a cycle. If the dominant cycle were 18 bars, for example, then the detuning of the filter would produce a 3-bar lead. This leading function is not huge, but it is significant.

A convenient trigger line is included in the adaptive band-pass filter to signal the more highly likely buy and sell points. The trigger is compute

as 90 percent of the amplitude of the adaptive band-pass filter line and is delayed by one bar. While the line crossings occur after the peak of the band-pass filter, phase lead provides for the generation of a timely signal. Significant trading signals should also include the criteria that the line crossing occur at greater than the +0.7 and less than the -0.7 reference lines.

Code Listing 11-4. EasyLanguage Code to Compute the Adaptive Band-Pass Filter

```
{
    Adaptive BandPass Indicator
    (c) 2013 John F. Ehlers
}
```

Inputs:

```
    Bandwidth(.3);
```

Vars:

```
    AvgLength(3),
    M(0),
    N(0),
    X(0),
    Y(0),
    alpha1(0),
    HP(0),
    a1(0),
    b1(0),
    c1(0),
    c2(0),
    c3(0),
    Filt(0),
    Lag(0),
    count(0),
    Sx(0),
    Sy(0),
    Sxx(0),
    Syy(0),
    Sxy(0),
    Period(0),
    Sp(0),
```

(Continued)

```

    Spx(0),
    MaxPwr(0),
    DominantCycle(0);

Arrays:
    Corr[48](0),
    CosinePart[48](0),
    SinePart[48](0),
    SqSum[48](0),
    R[48, 2](0),
    Pwr[48](0);

//Highpass filter cyclic components whose periods are
shorter than 48 bars
alpha1 = (Cosine(.707*360 / 48) + Sine (.707*360 / 48) - 1) /
Cosine(.707*360 / 48);
HP = (1 - alpha1 / 2)*(1 - alpha1 / 2)*(Close - 2*Close[1] +
Close[2]) + 2*(1 - alpha1)*HP[1] - (1 - alpha1)*
(1 - alpha1)*HP[2];

//Smooth with a Super Smoother Filter from equation 3-3
a1 = expvalue(-1.414*3.14159 / 10);
b1 = 2*a1*Cosine(1.414*180 / 10);
c2 = b1;
c3 = -a1*a1;
c1 = 1 - c2 - c3;
Filt = c1*(HP + HP[1]) / 2 + c2*Filt[1] + c3*Filt[2];
//Pearson correlation for each value of lag
For Lag = 0 to 48 Begin
    //Set the averaging length as M
    M = AvgLength;
    If AvgLength = 0 Then M = Lag;
    Sx = 0;
    Sy = 0;
    Sxx = 0;
    Syy = 0;
    Sxy = 0;
    For count = 0 to M - 1 Begin
        X = Filt[count];

```

Code Listing 11-4. (Continued)

```
Y = Filt[Lag + count];
Sx = Sx + X;
Sy = Sy + Y;
Sxx = Sxx + X*X;
Sxy = Sxy + X*Y;
Syy = Syy + Y*Y;
End;
If (M*Sxx - Sx*Sx)*(M*Syy - Sy*Sy) > 0 Then Corr[Lag] =
(M*Sxy - Sx*Sy)/SquareRoot((M*Sxx - Sx*Sx)*(M*Syy - Sy*Sy));
End;

For Period = 10 to 48 Begin
  CosinePart[Period] = 0;
  SinePart[Period] = 0;
  For N = 3 to 48 Begin
    CosinePart[Period] = CosinePart[Period] +
      Corr[N]*Cosine(360*N / Period);
    SinePart[Period] = SinePart[Period] + Corr[N]*Sine(360*N /
      Period);
  End;
  SqSum[Period] = CosinePart[Period]*CosinePart[Period] +
    SinePart[Period]*SinePart[Period];
End;

For Period = 10 to 48 Begin
  R[Period, 2] = R[Period, 1];
  R[Period, 1] = .2*SqSum[Period]*SqSum[Period] +
    .8*R[Period, 2];
End;

//Find Maximum Power Level for Normalization
MaxPwr = .995*MaxPwr;
For Period = 10 to 48 Begin
  If R[Period, 1] > MaxPwr Then MaxPwr = R[Period, 1];
End;
For Period = 3 to 48 Begin
  Pwr[Period] = R[Period, 1] / MaxPwr;
End;

//Compute the dominant cycle using the CG of the spectrum
Spx = 0;
Sp = 0;
```

(Continued)

Code Listing 11-4. (Continued)

```
For Period = 10 to 48 Begin
    If Pwr[Period] >= .5 Then Begin
        Spx = Spx + Period*Pwr[Period];
        Sp = Sp + Pwr[Period];
    End;
End;

If Sp <> 0 Then DominantCycle = Spx / Sp;
If DominantCycle < 10 Then DominantCycle = 10;

//Adaptive BandPass indicator tunes a BandPass filter to 90%
of the period of the Dominant Cycle
Vars:
    gammal(0),
    alpha2(0),
    beta1(0),
    BP(0),
    Peak(0),
    Signal(0),
    Lead(0),
    LeadPeak(0),
    LeadSignal(0);

beta1 = Cosine(360 / (.9*DominantCycle));
gammal = 1 / Cosine(360*Bandwidth / (.9*DominantCycle));
alpha2 = gammal - SquareRoot(gammal*gammal - 1);
BP = .5*(1 - alpha2)*(Filt - Filt[2]) + beta1*(1 +
alpha2)*BP[1] - alpha2*BP[2];
If Currentbar = 1 or CurrentBar = 2 Then BP = 0;

Peak = .991*Peak[1];
If AbsValue(BP) > Peak Then Peak = AbsValue(BP);
If Peak <> 0 Then Signal = BP / Peak;
Lead = 1.3*(Signal + Signal[1] - Signal[2] - Signal[3]) / 4;
LeadPeak = .93*LeadPeak[1];
If AbsValue(Lead) > LeadPeak Then LeadPeak = Absvalue(Lead);
If LeadPeak <> 0 Then LeadSignal = .7*Lead / LeadPeak;

Plot1(Signal);
Plot14(.9*Signal[1]);
Plot2(0);
Plot6(.707);
Plot10(-.707);
```

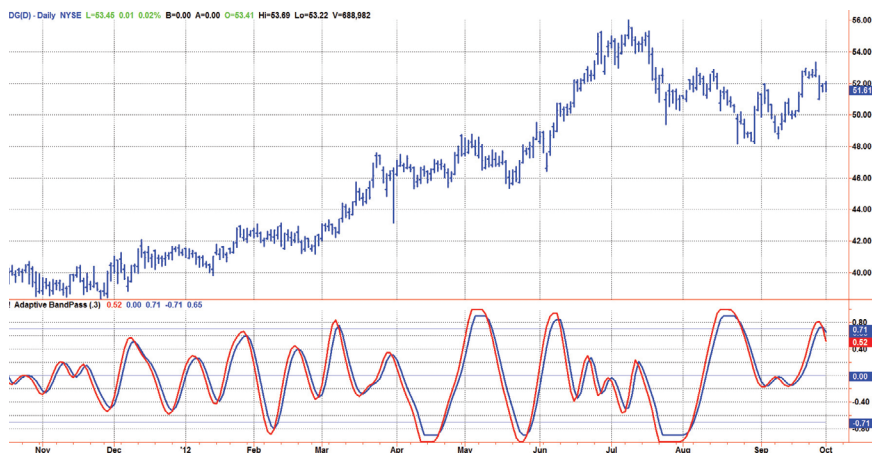


FIGURE 11.4 The Adaptive Band-Pass Filter Shows Buying and Selling Opportunities

The adaptive band-pass filter has been applied to Dollar General (symbol DG) over roughly the past year in Figure 11.4. This indicator accurately displays cyclic entry points when the two indicator lines cross outside the plus or minus 0.7 reference lines.

■ Adaptive Indicator Comparison

It is instructive to view the adaptive indicators on a single chart so they can be compared on a head-to-head basis. This comparison is available in Figure 11.5. The most striking difference is that the band-pass filter has more high-

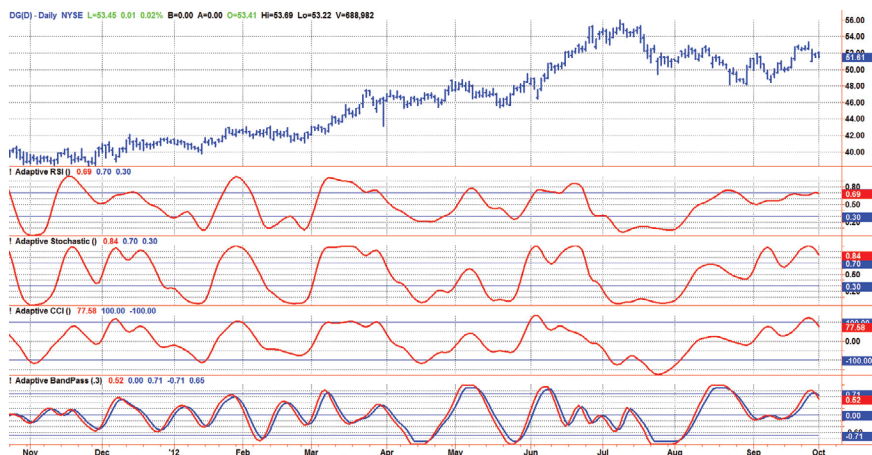


FIGURE 11.5 Comparison of the Adaptive Indicators

frequency components. It is also apparent to me that the RSI, Stochastic, and CCI all pretty much indicate the same thing in the broad general sense. Without indicator transforms to enhance interpretation, which we cover in Chapter 15, there is no overwhelming reason to select one indicator over another.

■ Key Points to Remember

1. When length is an input parameter to any indicator, that indicator can often be optimized by tuning that length to the dominant cycle period or a fraction thereof.
2. The preferred method of computing the dominant cycle is using the autocorrelation periodogram.
3. The RSI indicator is optimized over half the period of the measured dominant cycle to reach full-scale swings without oversaturation.
4. The stochastic indicator is optimized over the full period of the measured dominant cycle to ensure the highest close and the lowest close have been included.
5. The CCI indicator is optimized over the full period of the measured dominant cycle to ensure a complete measurement of the standard deviation.
6. The band-pass filter is optimized by tuning the filter to 90 percent of the measured dominant cycle period. This provides an approximate 60-degree phase lead in the indicator.

Notes

1. P. J. Kaufman, *Trading Systems and Methods*, 3rd ed. (New York: John Wiley & Sons, 1998), pp. 436–438.
2. T. S. Chande and S. Kroll, *The New Technical Trader* (New York: John Wiley & Sons, 1994).
3. J. Welles Wilder, Jr., *New Concepts in Technical Trading Systems* (Winston-Salem, NC: Hunter, 1978).
4. D. Lambert, *Commodities Magazine*, October 1980.